Michael Ichikawa

AI ENGINEER · AGENTIC SYSTEMS

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SUMMARY

As the fractional lead AI engineer for a small manufacturer, I own its software and AI, from architecture and the cloud platform to security and operations, starting with an LLM quoting app. Before that I shipped a production AI agent solo, from an empty repo to listings on the Microsoft Commercial Marketplace and the Teams Store, and built a multi-agent workflow engine from scratch. Earlier I earned an MS in Mathematics, taught college math for eight years, and spent three years in chip design at Intel.

EXPERIENCE

Lead AI Engineer (Fractional)

2026 – present

Small manufacturing company (client confidential) · Remote · Contract

- Own the company's software and AI as its sole engineer, reporting to the owners: architecture, development, deployment, security and operations
- Architect and run the company's Azure platform: a self-hosted AI assistant, staging and live environments per app, rollback, managed identity with no stored secrets, private networking, and CI/CD that blocks any change from merging or deploying until its tests pass
- Develop an LLM quoting app that extracts requirements from a customer's whole request (emails, documents, drawings, scans and call notes) in seconds and returns product model numbers, each value linked to its source line. The owners say finding that information can take a salesperson half a day
- Separate extraction from decisions. The LLM only extracts and quotes its sources word for word, deterministic rules code built from the company's quoting guide makes every product decision, and the app stops for human review on missing or conflicting information
- Let the owners change quoting rules in natural language, regression-tested against every open and approved quote and 50,000 synthetic requests, with a diff of what would change and approval by a second person

Founder & AI Engineer

2026

Microclaw LLC · Portland, OR · Marketplace listing · Code and evals

- Built Microclaw, one AI agent across Microsoft 365: 87 function-calling tools, a tool-calling loop that chains them until a task is complete, RAG semantic memory, multi-layer failure recovery, and model routing that escalates to a larger model only when a turn needs it
- Redesigned how the agent picks its tools. On 362 blind-written test requests it offered every tool a request needed 98.1 percent of the time, up from 87.0, with 49 percent fewer prompt tokens than sending every tool
- Took it from first commit to a Marketplace listing in about ten weeks, solo, through Marketplace certification and the Fulfillment and Metering APIs that make it billable, then Teams Store validation after seven review rounds
- Isolated tenants with Postgres row-level security: FORCE policies on every tenant-scoped table, an app role that cannot bypass them, and delegated Entra auth

TypeScript · Azure OpenAI · Microsoft Graph · Azure Postgres (RLS) · Bicep

Career Break · Full-Time Parent, Retraining into Software

2023 – 2026

Portland, OR

- Three years as primary caregiver
- Started in data engineering and ML, then moved to agentic development

Mathematics Faculty

2015 – 2023

Santa Rosa Junior College · Santa Rosa, CA

- Taught mathematics for eight years at an open-enrollment college: Algebra through Calculus III, Linear Algebra, Differential Equations and introductory Statistics

- Wrote my own lectures and exams, and my own version of an open-source precalculus textbook in LaTeX, reorganized by concept so trig functions came in week 2 instead of week 10

Physical Design Engineer

2008 – 2011

Intel Corporation · Semiconductor R&D

- Owned the layout of whole blocks in the Ivy Bridge (22nm) microprocessor, keeping timing, power and design-rule checks passing as changes landed in a large, version-controlled shared design
- Wrote Python and Perl automation for physical verification

PROJECTS

Multi-Agent Workflow Engine

mtichikawa.github.io/workflow-engine

A Python engine, built from scratch on the standard library, that turns a plain-English request into a graph of single-purpose agents. Every workflow is statically validated before it runs, and any step that acts outside the engine must sit behind a human approval gate. A learning loop infers a team's unwritten rules from their corrections, with no fine-tuning: on 150 blind-written messages, routing accuracy rose from 59 to 97 percent.

Python · agentic orchestration · MCP (client + server) · static graph validation

Terminal-Bench 3 Task

github.com/mtichikawa/klavis-tb3-submission

An original task for the Terminal-Bench 3 agent benchmark. Two exports of the same orders disagree because of four overlapping defects, and the agent has to recover the right values and work out how many separate causes there are. Claude Code on Claude Opus 5 and Codex on GPT-5.6 failed all six trials. Every agent got the numbers right and then reported the wrong number of causes.

TECHNICAL SKILLS

AI-Native Dev	Claude Code · directing AI agents to build, test and debug features, verified by tests and evals
Agents / LLM	agent orchestration · tool-calling agent loops · function calling · tool retrieval (KNN) · RAG (retrieval-augmented generation) · vector search & embeddings (pgvector · FAISS · BigQuery Vector Search) · semantic memory · multi-agent workflows · document extraction with source citations · prompt engineering · human-in-the-loop gating · MCP (client + server) · in-context learning
Reliability & Cost	failure-mode engineering (backoff/retry · context compaction · loop detection · abort ceilings) · model routing with mid-task escalation · input-token reduction · multi-tenant isolation (Postgres RLS)
Evaluation	LLM evaluation · eval design · expected-facts scoring · A/B against baselines · blind held-out test sets · pytest · Vitest · GitHub Actions CI
Built with	Python · TypeScript · Node.js · SQL · Azure OpenAI · Claude API (Anthropic) · Microsoft Graph · PostgreSQL · BigQuery · pandas · scikit-learn · Azure (Container Apps · Entra · managed identity · Bicep) · AWS (S3 · Lambda) · GCP (Vertex AI · Cloud Run) · Docker · FastAPI / REST

EDUCATION

MS Mathematics — The City College of New York — 2014

BS Mechanical Engineering — University of California, Berkeley — 2008